

Cross-Region VPC Peering POC

Mumbai (ap-south-1) to Singapore (ap-southeast-1). Goal: an application server reaches a private database server over private IPs, with no internet exposure for the database.

1. Architecture

	App VPC	DB VPC
Name	app-vpc01	Db-Vpc
Region	Mumbai	Singapore
VPC ID	vpc-076a897c96607b5db	vpc-023cc299cb387fb82
CIDR	10.0.0.0/24	10.20.0.0/16
Subnet	app-public-subnet01 , 10.0.0.0/26 , ap-south-1a	db-private-subnet , ap-southeast-1c
Internet Gateway	app-vpc-igw	None
Server	poc-app-server (public IP)	DB server (no public IP)

Peering connection: pcx-0f983a7a9d667cc8f (requester app-vpc01 , acceptor Db-Vpc).

Internet -> [App server, Mumbai] ==peering (private)==> [DB server, Singapore, no int



2. App VPC (Mumbai)

One VPC, one public subnet, an Internet Gateway, and a route table with 0.0.0.0/0 to the gateway.

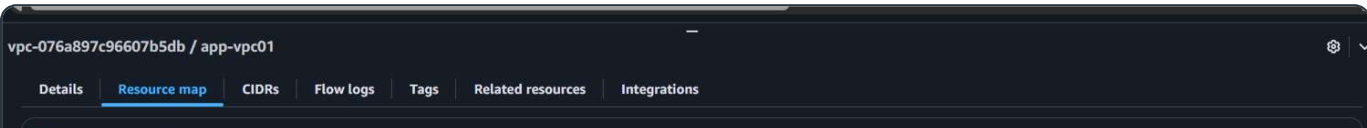




Figure 1. App VPC resource map: `app-vpc01` with its public subnet, route table `app-Rt` (3 routes including local) and Internet Gateway `app-vpc-igw`.

YOUR VPCS (1/5)

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public Access	IPv4 CIDR	IPv6 CIDR
-	<code>vpc-0fad95f0785304815</code>	Available	-	-	Off	<code>172.31.0.0/16</code>	-
My-VPC	<code>vpc-044317c6a349b141f</code>	Available	-	-	Off	<code>10.0.0.0/16</code>	-
app-vpc01	<code>vpc-076a897c96607b5db</code>	Available	-	-	Off	<code>10.0.0.0/24</code>	-

vpc-076a897c96607b5db / app-vpc01

Details

VPC ID <code>vpc-076a897c96607b5db</code>	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set <code>dopt-07c0ac254ea45220d</code>	Main route table <code>rtb-028c668f6ca4416df</code>
Main network ACL <code>acl-082ed80a2af2ea5ea</code>	Default VPC No	IPv4 CIDR <code>10.0.0.0/24</code>	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID <code>533003975261</code>
Encryption control ID -	Encryption control mode -		

Figure 2. App VPC details: IPv4 CIDR `10.0.0.0/24`, DNS resolution enabled, default tenancy.

Subnets (1/8)

Name	Subnet ID	State	VPC	Block Public A...	IPv4 CIDR	IPv6 CIDR
Private-subnet-1	<code>subnet-04ee77c5fca2720bf</code>	Available	<code>vpc-044317c6a349b141f My-...</code>	Off	<code>10.0.3.0/24</code>	-
app-public-subnet01	<code>subnet-088f07ab1532b17f0</code>	Available	<code>vpc-076a897c96607b5db app-...</code>	Off	<code>10.0.0.0/26</code>	-
Public-subnet-2	<code>subnet-049c49c2bf8941806</code>	Available	<code>vpc-044317c6a349b141f My-...</code>	Off	<code>10.0.2.0/24</code>	-
Private-subnet-2	<code>subnet-0e4143a5329bb5059</code>	Available	<code>vpc-044317c6a349b141f My-...</code>	Off	<code>10.0.4.0/24</code>	-
-	<code>subnet-0ae3fbdcb0a1f56e7</code>	Available	<code>vpc-0fad95f0785304815</code>	Off	<code>172.31.16.0/20</code>	-
Public-subnet-1	<code>subnet-029102d7b137fc84f</code>	Available	<code>vpc-044317c6a349b141f My-...</code>	Off	<code>10.0.1.0/24</code>	-

subnet-088f07ab1532b17f0 / app-public-subnet01

Details

Subnet ID <code>subnet-088f07ab1532b17f0</code>	Subnet ARN <code>arn:aws:ec2:ap-south-1:533003975261:subnet/subnet-088f07ab1532b17f0</code>	State Available	Block Public Access Off
IPv4 CIDR <code>10.0.0.0/26</code>	Available IPv4 addresses 59	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone <code>aps1-az1 (ap-south-1a)</code>	Network border group <code>ap-south-1</code>	VPC <code>vpc-076a897c96607b5db app-vpc01</code>	Route table <code>rtb-028c668f6ca4416df app-Rt</code>
Network ACL <code>acl-082ed80a2af2ea5ea</code>	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 -	Owner <code>533003975261</code>	

Figure 3. App subnet `app-public-subnet01`: `10.0.0.0/26` in `ap-south-1a`, linked to `app-Rt`.

3. DB VPC (Singapore)

A private VPC with one subnet and no Internet Gateway or NAT. Its CIDR must not overlap the App VPC.

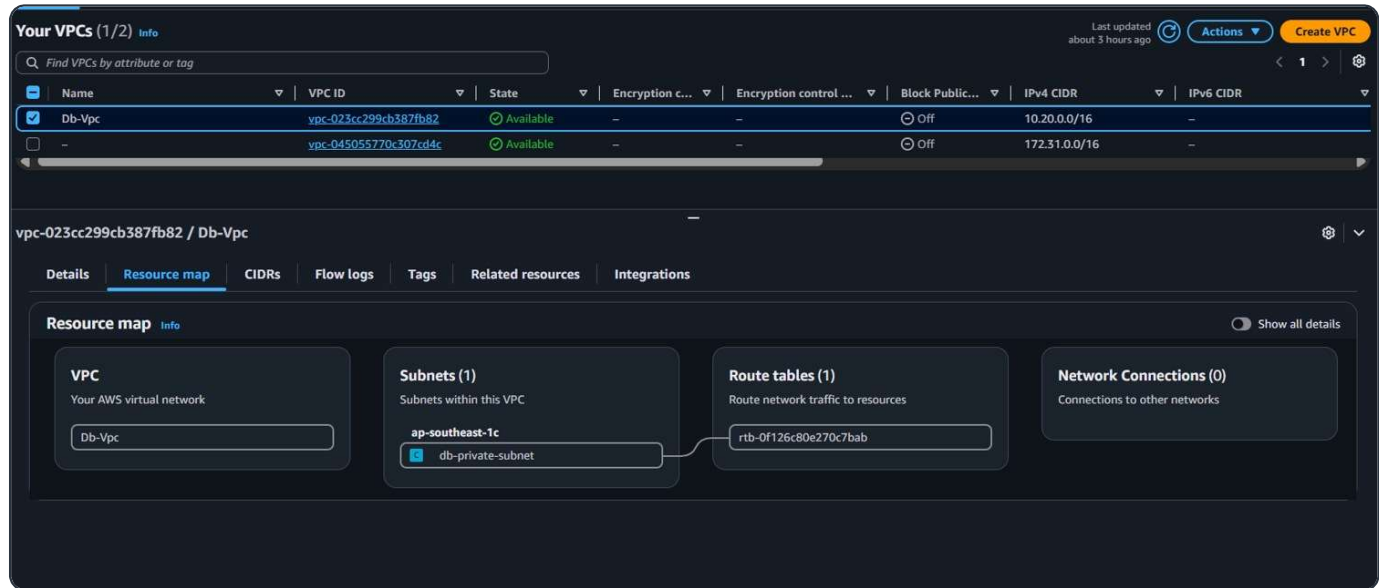


Figure 4. DB VPC in Singapore: Db-Vpc (10.20.0.0/16), one private subnet, one route table and no network connections (no Internet Gateway or NAT).

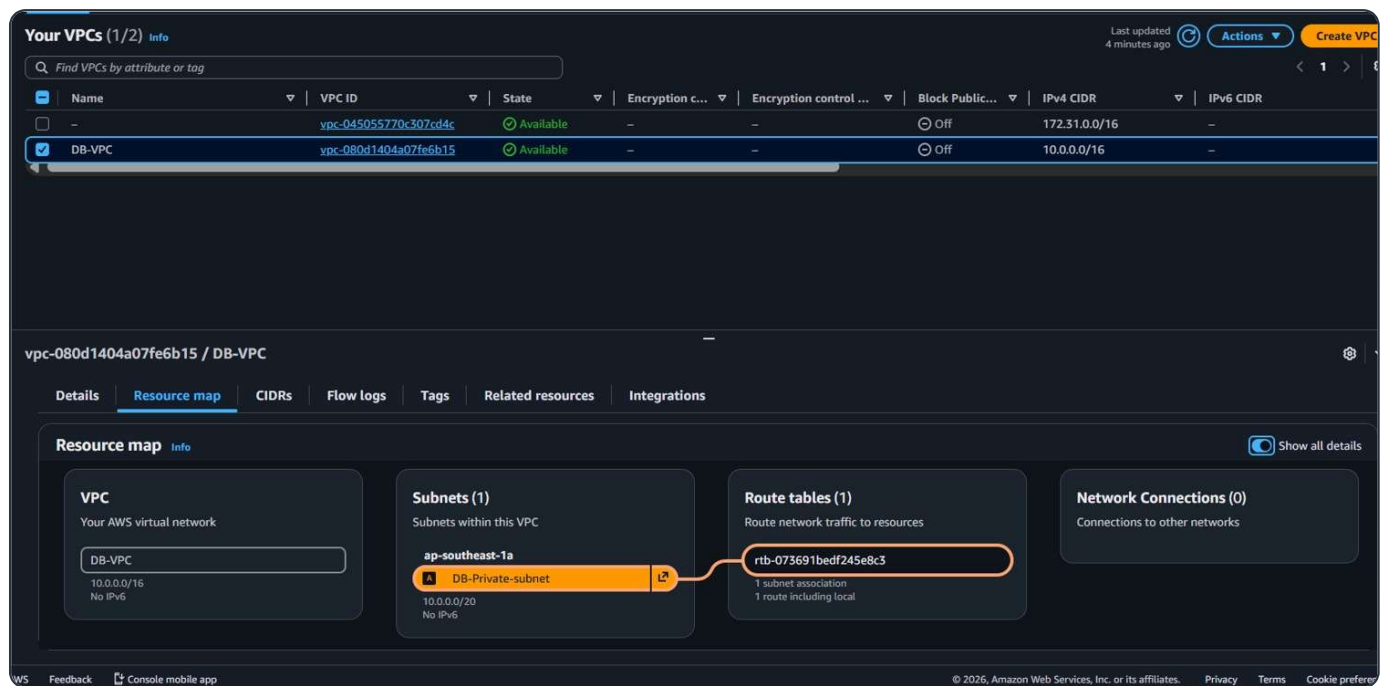


Figure 5. First attempt: DB-VPC was created as 10.0.0.0/16, which overlaps the App VPC (10.0.0.0/24). It was deleted and recreated.

4. Peering and routes

The request was created in Mumbai (another region, accepter VPC ID pasted) and accepted in Singapore. Active status only means the link exists, so each side also needs a route to the other VPC's CIDR.

Route table	Destination	Target
app-Rt (Mumbai)	10.20.0.0/16	pcx-0f983a7a9d667cc8f
DB route table (Singapore)	10.0.0.0/24	pcx-0f983a7a9d667cc8f

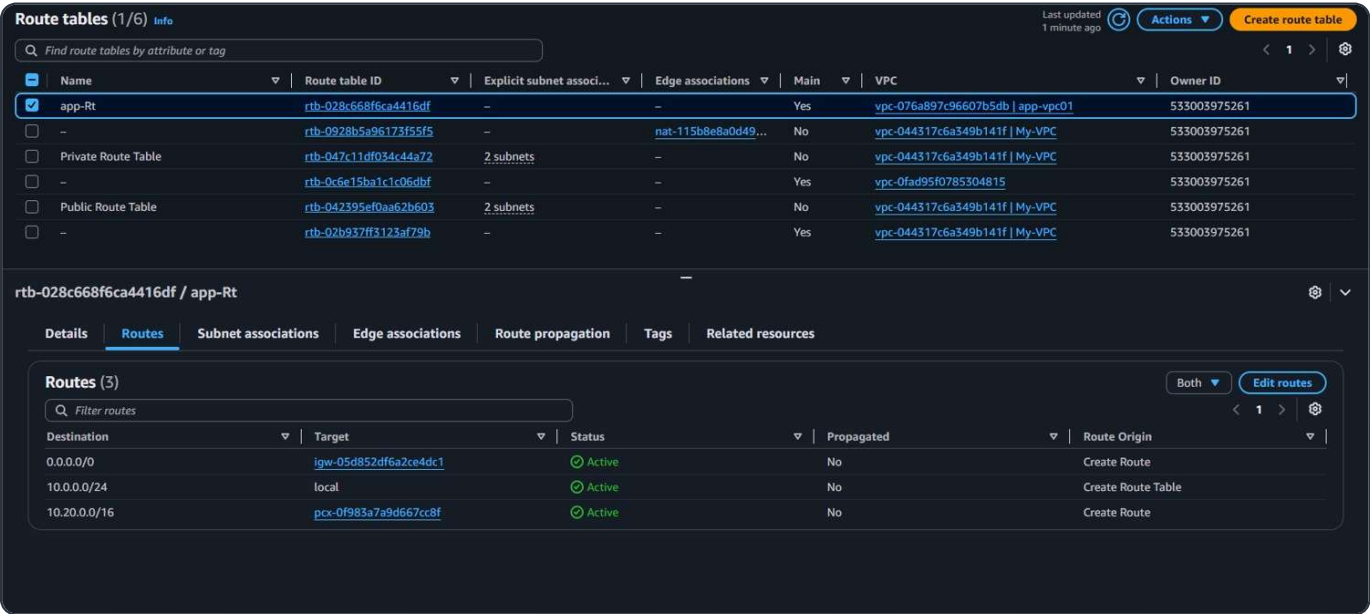
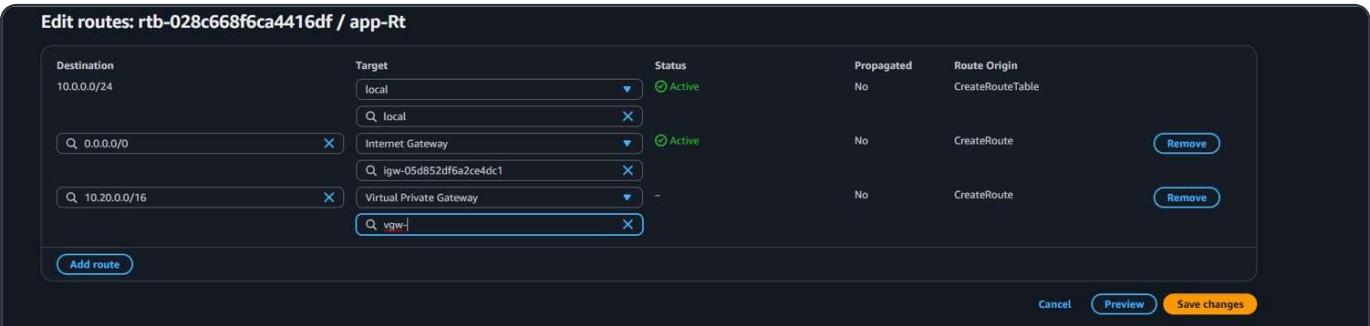


Figure 6. Mumbai route table app-Rt : internet route, local route and 10.20.0.0/16 to the peering connection.



Figure 7. Singapore route table: 10.0.0.0/24 to the same peering connection, plus the local route.



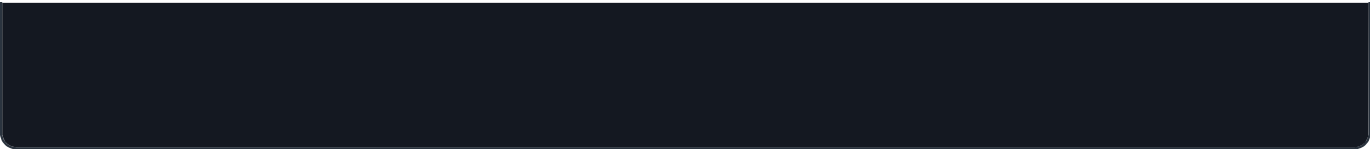


Figure 8. Mistake while adding the route: target type was set to Virtual Private Gateway. It must be Peering Connection.

5. DB security group

Only the App VPC range is allowed. A CIDR is used instead of a security group reference because security groups cannot be referenced across regions.

Type | Info

ssh

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Custom

Source | Info

Add CIDR, prefix list or security group

10.0.0.0/24 X

Description - optional | Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 8080, 10.0.0.0/24)

Remove

Type | Info

Custom TCP

Protocol | Info

TCP

Port range | Info

8080

Source type | Info

Custom

Source | Info

Add CIDR, prefix list or security group

10.0.0.0/24 X

Description - optional | Info

e.g. SSH for admin desktop

▼ Security group rule 3 (ICMP, All, 10.0.0.0/24)

Remove

Type | Info

Custom ICMP - IPv4

Protocol | Info

All

Port range | Info

All

Source type | Info

Custom

Source | Info

Add CIDR, prefix list or security group

10.0.0.0/24 X

Description - optional | Info

e.g. SSH for admin desktop

Figure 9. DB security group rules: SSH 22, TCP 8080 and ICMP, all from 10.0.0.0/24 .

The DB server was launched with this user-data script, which starts a test listener on port 8080:

```
#!/bin/bash
nohup python3 -m http.server 8080 --directory /tmp &
```

6. Servers

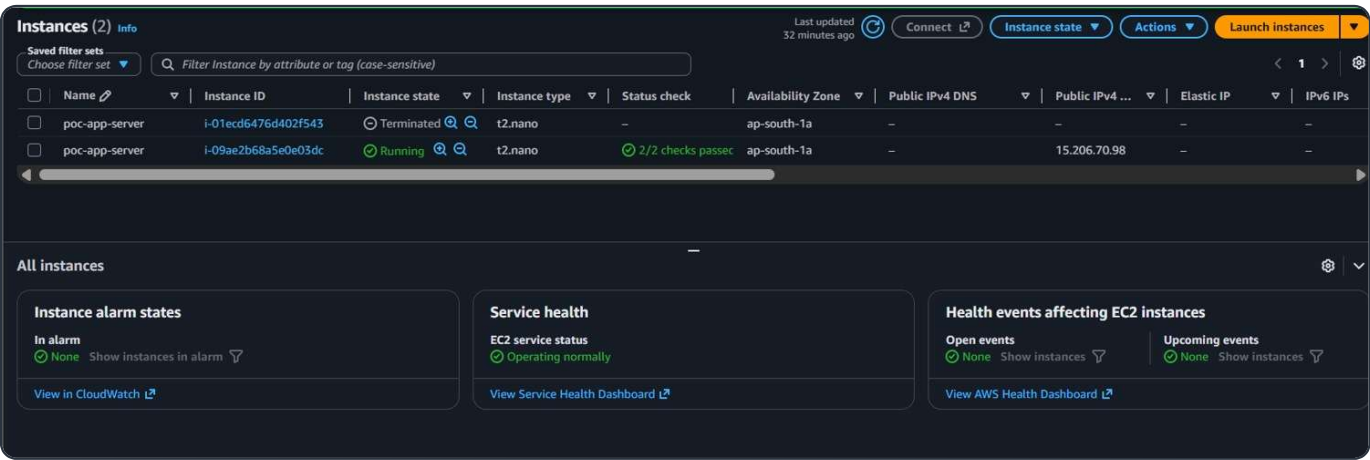


Figure 10. Mumbai EC2 console: poc-app-server running with 2/2 checks passed (an earlier instance shows as terminated).

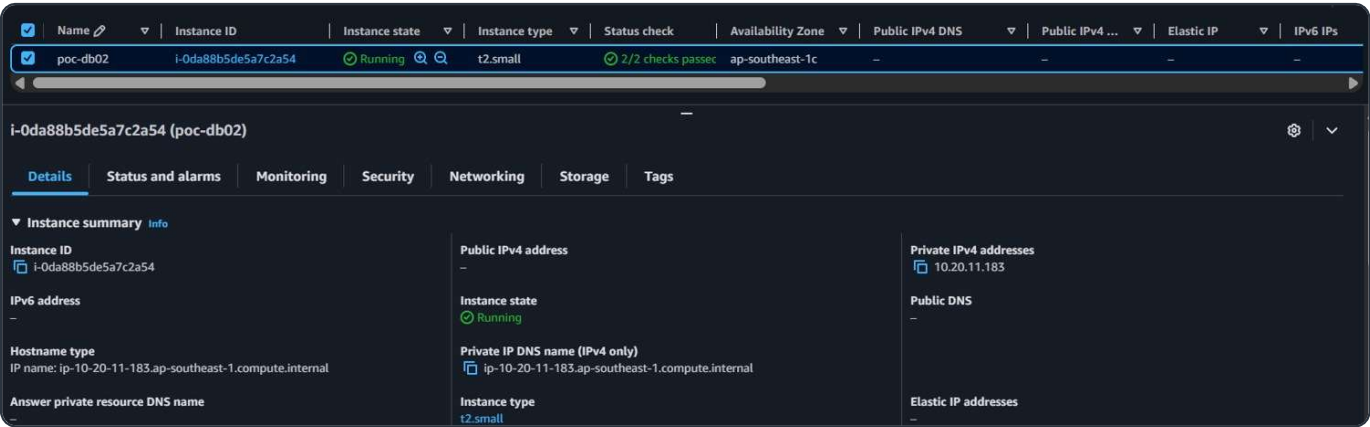


Figure 11. Singapore EC2 console: poc-db02 running in ap-southeast-1c with private IP 10.20.11.183 and no public IP.

7. Test results

Test	Command (from App server)	Result
Ping	ping 10.20.11.183	Replies, about 60 ms
HTTP	curl http://10.20.6.2:8080	Directory listing returned
SSH	ssh -i db-key.pem ec2-user@10.20.6.2	Logged in to the DB server
File copy	scp ... test.txt ec2-user@10.20.6.2:/tmp/	100%, then curl ../test.txt returned the text

Servers were relaunched during the POC, so two pairs appear: 10.0.0.34 to 10.20.11.183 ,
https://claude.ai/chat/a5091c6f-b3ab-437d-909b-39da4b21512a?artifact=fcbbc1e5-82d8-480c-89ae-8adb53630ead

```
[ec2-user@ip-10-0-0-34 ~]$ mkdir poc-folder
[ec2-user@ip-10-0-0-34 ~]$ echo "hello from App server" > poc-folder/test.txt
[ec2-user@ip-10-0-0-34 ~]$ scp -i db-key.pem -r poc-folder ec2-user@10.20.11.183:/home/ec2-user/
Warning: Identity file db-key.pem not accessible: No such file or directory.
The authenticity of host '10.20.11.183 (10.20.11.183)' can't be established.
ECDSA key fingerprint is SHA256:9A2gJTKRye1Ru4vE0BAsZV6MeKqTvc8jPEBEvgHxbo.
Your key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? ^Z
[1]+  Stopped                  scp -i db-key.pem -r poc-folder ec2-user@10.20.11.183:/home/ec2-user/
[ec2-user@ip-10-0-0-34 ~]$ ping 10.20.11.183
PING 10.20.11.183 (10.20.11.183) 56(84) bytes of data.
64 bytes from 10.20.11.183: icmp_seq=1 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=2 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=3 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=4 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=5 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=6 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=7 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=8 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=9 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=10 ttl=127 time=61.0 ms
64 bytes from 10.20.11.183: icmp_seq=11 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=12 ttl=127 time=60.9 ms
64 bytes from 10.20.11.183: icmp_seq=13 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=14 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=15 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=16 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=17 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=18 ttl=127 time=60.9 ms
64 bytes from 10.20.11.183: icmp_seq=19 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=20 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=21 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=22 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=23 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=24 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=25 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=26 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=27 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=28 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=29 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=30 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=31 ttl=127 time=60.8 ms
64 bytes from 10.20.11.183: icmp_seq=32 ttl=127 time=60.6 ms
64 bytes from 10.20.11.183: icmp_seq=33 ttl=127 time=60.7 ms
64 bytes from 10.20.11.183: icmp_seq=34 ttl=127 time=60.6 ms
^Z
[2]+  Stopped                  ping 10.20.11.183
[ec2-user@ip-10-0-0-34 ~]$
```

i-09ae2b68a5e0e03dc (poc-app-server)

```

      #
    ____#____
   /#####\
  /         \
 ~~~~~~     | http://aws.amazon.com/linux/amazon-linux-2023
  V~/       ->
   /m/\_____/_/
  /          _/
 /            \_

Last login: Mon Oct  9 07:48:51 2026 from 13.233.177.5
[ec2-user@ip-10-0-0-21 ~]$ ls
db-key.pem
[ec2-user@ip-10-0-0-21 ~]$ ssh -i db-key.pem ec2-user@10.20.6.2
Load key "db-key.pem": invalid format
ec2-user@10.20.6.2: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-0-21 ~]$ ss -f db-key.pem
[ec2-user@ip-10-0-0-21 ~]$ ^C
[ec2-user@ip-10-0-0-21 ~]$ ls -l db-key.pem
-rw-r--r-- 1 ec2-user ec2-user 1678 Oct  9 09:10 db-key.pem
[ec2-user@ip-10-0-0-21 ~]$ chmod 400 db-key.pem
[ec2-user@ip-10-0-0-21 ~]$ ssh-keygen -y -f db-key.pem
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQBgVclNDoc3nATdfeVtCSFBUft9KHqO2IBoR6qugN+ILAvrdzW7814wtIDxH9S5CFRjXCYNpacegcCNqdAeJ8BajfvW3DXkJKXjk7SzOVRT9PbgIMMO/pieqe8wnjNBW5yd12ce3AJCeFrUBjBV860AAPGuaKOCxOSw+4GJE7d/6imwscWOPZICvhhyLoznfmLn9toKQBd/+S8S2ooolk/+agkwFMV/EINYt+XBvEybu4obGB42xnYXgKgBFkmJUhpTzVeelyIQPeSP06A1+QNdl3xCe9x3K7I/Q9gen42fmaTZkdPlLbwQvtgdh2b32ANaBHwaJSgv7gr+DmsFTLzl,
[ec2-user@ip-10-0-0-21 ~]$ ssh -i db-key.pem ec2-user@10.20.6.2

      #
    ____#____
   /#####\
  /         \
 ~~~~~~     | http://aws.amazon.com/linux/amazon-linux-2023
  V~/       ->
   /m/\_____/_/
  /          _/
 /            \_

[ec2-user@ip-10-20-6-2 ~]$
```

```
~\#####  
~~ \####/  
~~ #####  
~~ \|      https://aws.amazon.com/linux/amazon-linux-2023  
~~ v-'__>  
  
_/_/_/_____  
m/'          |  
[ec2-user@ip-10-20-6-2 ~]$ exit  
logout  
Connection to 10.20.6.2 closed.  
[ec2-user@ip-10-0-0-21 ~]$ mkdir poc-folder  
[ec2-user@ip-10-0-0-21 ~]$ echo "hello from App server" > poc-folder/test.txt  
[ec2-user@ip-10-0-0-21 ~]$ ssh -i db-key.pem ec2-user@10.20.6.2 "cat poc-folder/test.txt"  
cat poc-folder/test.txt: No such file or directory  
[ec2-user@ip-10-0-0-21 ~]$ scp -i db-key.pem poc-folder/test.txt ec2-user@10.20.6.2:/tmp/  
test.txt                                                                100%   22    0.4KB/s   00:00  
[ec2-user@ip-10-0-0-21 ~]$ curl http://10.20.6.2:8080/test.txt  
Hello from App server  
[ec2-user@ip-10-0-0-21 ~]$ ssh -i db-key.pem ec2-user@10.20.6.2  
_# #  
~\ #### Amazon Linux 2023  
~~ \#####  
~~ #####  
~~ \|      https://aws.amazon.com/linux/amazon-linux-2023  
~~ v-'__>
```

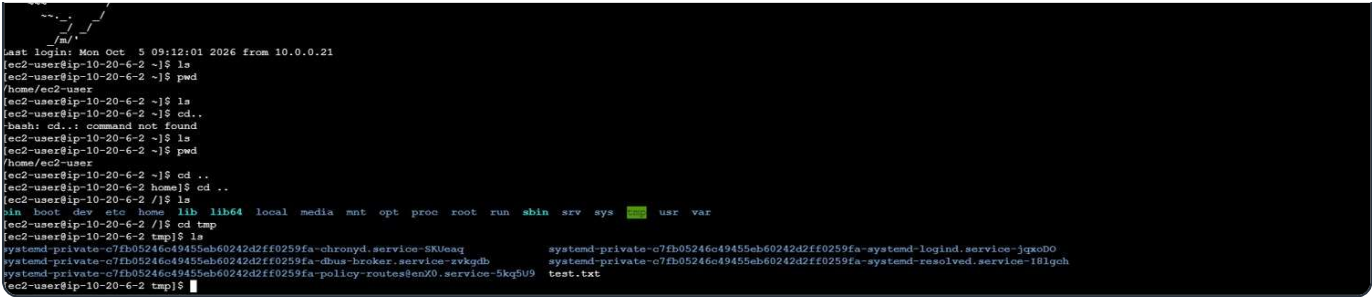


Figure 14. File transfer: `test.txt` copied with `scp`, read back with `curl` on port 8080, and found in `/tmp` on the DB server.

8. Problems faced and fixes

Problem	Cause	Fix
First DB VPC was 10.0.0.0/16	Overlaps the App VPC, so peering is impossible	A primary CIDR cannot be edited. Recreated as 10.20.0.0/16
Route target was Virtual Private Gateway	Wrong target type	Changed to Peering Connection
App subnet auto-assign public IP is No	Subnet setting not enabled	Public IP enabled at instance launch
Load key: invalid format	Pasting the key text damaged its line breaks	Copied the original .pem with <code>scp</code> ; <code>ssh-keygen -y -f</code> then worked
scp: dest open ... Permission denied	Old read-only copy of the key was in the way	Removed the old file and copied again

9. Key points